

I Tried Looking for Prime-Number Patterns in Language. The Interesting Part Is What Didn't Happen.

A short introduction to *Riemann-Resonant Linguistics*, a new pilot paper about language, structure, null models, and the fine art of not fooling oneself.

There are certain phrases one should approach with protective gloves.

"Prime numbers in language" is one of them.

Not because it is obviously nonsense. That would be too easy, and frankly less entertaining. The danger is subtler: prime numbers are beautiful, language is mysterious, and human beings are disastrously good at finding patterns in wallpaper, clouds, markets, scripture, static, toast, and occasionally each other.

So when I began working on a paper called **Riemann-Resonant Linguistics**, the first problem was not mathematical.

It was hygienic.

How does one ask whether language has measurable prime-indexed positional structure without accidentally building a machine for numerology with a command-line interface?

Darlings, this is how.

You build the machine so that it is allowed to say **no**.

And, rather satisfyingly, it did.

What the paper is actually about

The paper introduces **RRLANG**, a small Rust-based research instrument for measuring positional structure in language corpora.

That sounds terribly grand, so let us translate.

Most language statistics ask questions like:

- How often does this word appear?
- How many letters, sounds, or symbols does this language use?
- How common is this grammatical pattern?

- How predictable is the next character or word?

These are useful questions. They are the respectable shoes of quantitative linguistics: sturdy, well-polished, and invited to conferences.

RRLANG asks a slightly different question:

Not just **what** appears in a text, but **where** it appears.

For example, suppose we mark every word boundary in a text. That gives us a map of where words begin and end. Or suppose we mark every vowel, punctuation mark, high-frequency character, or repeated class of symbol. We can then ask:

- How far apart are these events?
- Are the gaps regular or irregular?
- Do the events cluster?
- Do they survive comparison against shuffled versions of the same text?
- Do they show unusual behaviour at prime-numbered positions?
- Do they remain unusual once local patterns are preserved by stronger null models?

That last phrase, **null models**, is the grown-up at the party.

A null model is a deliberately boring comparison version of the data. It preserves some features while destroying others. For instance, a shuffled version of a text might preserve the number of vowels but destroy their original order. A Markov model might preserve local character habits, such as which characters tend to follow which, while destroying larger-scale structure.

So the question becomes:

Does this measured structure still look unusual after we compare it to many carefully controlled boring alternatives?

If yes, perhaps something interesting is happening.

If no, we do not pout. We learn.

Why prime numbers are involved at all

Prime numbers are numbers divisible only by themselves and one: 2, 3, 5, 7, 11, 13, and so on. They have fascinated mathematicians for centuries because they are simple to define and fiendishly difficult to predict in detail.

The paper does **not** claim that prime numbers cause language.

It does **not** claim that language solves the Riemann Hypothesis.

It does **not** claim that a text can be declared human, alien, divine, artificial, or “spiritually resonant” because something peculiar happened around the number 37. We are not doing séance mathematics in a velvet hat.

The point is more modest.

Prime-numbered positions can be used as a structured probe. They give us a way to sample positions in a sequence that is neither simply every second symbol nor every tenth symbol nor an arbitrary random draw. We can then ask whether certain linguistic events behave differently around prime-indexed positions or prime-sized gaps than they do under controlled null models.

That is all.

It is a measuring trick, not a metaphysical declaration.

The encoding ladder: because raw binary is a trap wearing eyeliner

One of the central safeguards in the paper is the **encoding ladder**.

A text can be represented in many ways:

- raw UTF-8 bytes,
- graphemes or written characters,
- character classes such as vowel, consonant, punctuation, or boundary,
- word-boundary maps,
- frequency classes,
- and, in future versions, phonemes, morphemes, syllables, syntax, and historical alignments.

This matters because raw binary data can be extremely misleading.

A Japanese text, an Arabic text, and an English text are not merely “different languages” when stored in a file. They may also use different byte patterns, different Unicode behaviours, different punctuation conventions, different spacing systems, and different script structures. If we naïvely convert everything to binary and become excited whenever a number sparkles, we are no longer doing linguistics. We are doing confetti.

So RRLANG v0.3 deliberately uses what the paper calls a **linguistic profile**. In the main comparisons, raw UTF-8 diagnostics are skipped. The focus is placed on more linguistically interpretable encodings such as:

- grapheme,
- grapheme class,
- word boundary,
- frequency class.

This is less flashy.

It is also much safer.

Science is often the art of making one’s own favourite idea less able to flatter oneself.

The Mesu question

The paper uses **Mesu**, a constructed language, as a case study.

A less careful version of this project might have tried to prove that Mesu was secretly special, unusually harmonious, mathematically resonant, or otherwise wearing a little crown.

That is not what happened.

The core multilingual test used a canonical set of **Universal Declaration of Human Rights** translations. This gives a parallel baseline: many languages expressing broadly comparable content.

RRLANG v0.3 ran across 17 canonical UDHR texts using the linguistic-profile encodings and 100 null samples.

Several natural-language translations retained Markov-2-surviving signals in metrics involving word boundaries, punctuation, vowels, gap entropy, run entropy, and prime-gap affinity.

Mesu did not.

Mesu UDHR had:

zero filtered Markov-2 survivors.

With punctuation allowed: zero.

With punctuation excluded: zero.

Quiet as a well-bred ghost.

At first glance, this may look disappointing. It is not.

It is one of the most important results in the paper.

Why? Because it shows that the instrument is not simply rewarding the author's own constructed-language case study. It is not a Mesu-flattery machine. It is capable of returning the answer: "No notable surviving signal here under these tests."

That matters.

A method that only confirms one's hopes is not a method. It is a mirror with delusions of peer review.

The register test: poetry did not rescue the drama

After the UDHR result, the obvious question was whether Mesu was quiet only because UDHR legal prose is, shall we say, not exactly intoxicated with aesthetic wildness.

So the next test compared Mesu across registers:

- Mesu UDHR,
- Mesu Dylan Thomas text,
- Mesu Basho text.

If Mesu's interesting structure lived in poetry or literary register rather than legal prose, this was where one might expect it to emerge.

It did not.

The Mesu register comparison produced no trusted Markov-2 survivors after filtering.

Again, this is not a failure. It is a boundary.

The paper can now say, honestly:

Under the current RRLANG v0.3 tests, Mesu does not show surviving prime-gap or positional-structure signals in either the canonical UDHR baseline or the tested literary/register samples.

That is not romantic.

It is useful.

And, as a general rule, "useful" survives longer than "romantic" in methods papers, marriages, software projects, and cookware.

The controls screamed, which is exactly what they were paid to do

A quiet Mesu result would be concerning if everything were quiet. Perhaps the instrument was too weak. Perhaps the filtering was too strict. Perhaps the whole apparatus was staring at language with the computational equivalent of a cold teaspoon.

So the paper also ran deterministic controls.

These included things like:

- duplicated Mesu text,
- punctuation-noise Mesu,
- random-bit style controls.

The controls produced strong survivor signals.

Some were huge. The strongest word-boundary entropy control signals had z-scores around -94, which is not a subtle cough from the back row. That is the data throwing a chair.

This is important because it shows two things at once:

1. RRLANG does **not** automatically produce positive claims for Mesu.
2. RRLANG **does** respond strongly when artificial positional structure is inserted.

That combination is the core pilot result.

The instrument is not dead.

The instrument is not sycophantic.

That is a surprisingly good start.

What the paper claims

The paper's main claim is deliberately modest:

Language can be studied as an ordered symbolic system, and positional metrics can be tested across controlled encodings and null models to see whether structure survives beyond frequency and local sequence habits.

In less formal terms:

Language is not just a bag of words. It is an ordered thing.

The same symbols in a different order can mean something entirely different. So there may be value in measuring placement, spacing, gaps, boundaries, and recurrence structures — not instead of traditional linguistic methods, but alongside them.

The paper presents RRLANG as a **pilot instrument**, not a finished theory of language.

It proposes a framework.

It demonstrates a pipeline.

It reports a clean negative result for Mesu.

It reports strong positive controls.

It argues for careful future work.

This is the academic equivalent of saying: the machine turns on, it does not immediately lie to us, and it reacts when poked with a sufficiently obvious stick.

For a first paper, that is not bad at all.

What the paper does not claim

This part matters, because without it the whole thing risks wandering into the glitter swamp.

The paper does **not** claim:

- that prime numbers cause language,
- that the Riemann Hypothesis is solved or advanced by language data,
- that raw binary structure is linguistic structure,
- that Mesu is proven special,
- that a single translation represents an entire language,
- that statistical anomaly automatically means meaning,
- that artificiality, authorship, or origin can be inferred from these metrics alone.

Those are not footnotes. They are load-bearing walls.

The method only becomes scientifically interesting if it refuses to overinterpret itself.

Why this is worth publishing, even as a pilot

The real value of this paper is not that it discovers some grand hidden musicality in Mesu.

It does something better.

It establishes a cautious experimental framework for asking a strange but legitimate question:

Can language systems have measurable positional signatures that survive strong comparison against null models?

That question sits near several existing fields:

- quantitative linguistics,
- corpus linguistics,
- information theory,
- stylometry,
- Markov models,
- sequence analysis,
- symbolic dynamics,
- computational linguistics,
- and, very carefully, prime-indexed or spectral analysis.

The paper does not pretend none of that previous work exists. It positions RRLANG as a small new instrument in that broader cabinet: a tool for examining event maps, gaps, boundaries, runs, and prime-indexed probes in language corpora.

The current evidence is not enough to make sweeping language-wide claims.

It is enough to justify the next version.

Future work should add:

- more languages,
- more corpora,
- larger null runs,
- multiple-comparison correction,
- true grapheme-cluster handling,
- better script-aware vowel and consonant classification,
- windowed corpus analysis,
- morphology-aware encodings,
- phonemic encodings,
- and cleaner publication-grade reproducibility manifests.

In other words: less mystique, more tables.

A tragic loss for theatre. A gain for science.

The most important result may be the negative one

There is a particular kind of intellectual honesty in building a tool to test a pet idea and then reporting that the pet did not perform tricks on command.

Mesu did not light up under the current trusted-survivor criterion.

The controls did.

That is the story.

And it is a much stronger story than “my constructed language is obviously special because I measured it with a thing I also invented.”

No, darlings.

The better version is:

I built a cautious instrument for measuring positional structure in language. It found structure in some natural-language baselines and in artificial controls. It did not falsely crown Mesu. Therefore, it may be worth developing further.

That is sober.

That is publishable.

That is, irritatingly enough, science.

Where to read the paper

The full paper, source materials, and repository are available here:

Repository and paper: [INSERT REPOSITORY LINK HERE]